AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 1 · TEACHER GUIDE

AI or Not?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Teach one connected idea.

Model the bus, transfer to umbrellas, guide the photo-library explanation, then reserve the greenhouse for independent evidence.

Keep one habit: Look for a mechanism, the deciding fact, and a defensible limit.

MAIN LESSON · SECTION 1

Prepare this week

> One learning aim, one main teaching plan, and a clear boundary between practice and independent evidence.

What learners must understand

Primary target: Distinguish an observed result from an explanation of how it was produced

Avoid making AI a yes-or-no property of a whole product before defining the component and decision. Modern **learned model**s and symbolic reasoning both belong in the broader AI discussion.

The learning goal here is **evidence**-based **mechanism** classification, not memorising a list of smart objects. Do not infer **training** from responsiveness, language, movement or one correct **output**.

By the end, I can separate a learned **prediction**, a **written rule** and a **human decision**.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- The bus diagram and photo-library case from the student chapter
- Blank paper and three colours of pencil
- A copy of workbook tasks W1 to W5; no online accounts or real student data

Main route: three sessions

Session	Sequence and minutes	Resources
Session 1 · Understand · 45 min	0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.	Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.
Session 2 · Practise · 45 min	0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.	Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

Session	Sequence and minutes	Resources
Session 3 · Investigate and transfer · 45 min	0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.	Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening question and what to listen for

A classroom clock speaks the time. Write one thing you observed and one thing you cannot determine just from hearing it.

You observed spoken time. Speech alone does not tell you whether the message was selected by a timetable, a learned system or a person.

▼ **Alternative timing options, repeated targets and vocabulary**

Teach the learning cycle

Retrieval answer guidance

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A bus displays a delay **prediction** learned from past journeys. A **written rule** turns the display red above 10 minutes. A driver can override the message. Identify all three **mechanism**s.

Would seeing a red display prove that learning occurred?

Concept to explain

Look for **evidence** of how the system reasons or learns, not just its label or screen.

Term	Meaning
AI system	a system using learned patterns or symbolic reasoning
fixed rule	an instruction written in advance
human-controlled	a person makes the important choice
evidence	a clue that supports or changes a claim

Prepare the class

This is an alternative compact allocation. The main starting plan above uses three 45-minute sessions; allow more time for the full investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

You observed spoken time. Speech alone does not tell you whether the message was selected by a timetable, a learned system or a person.

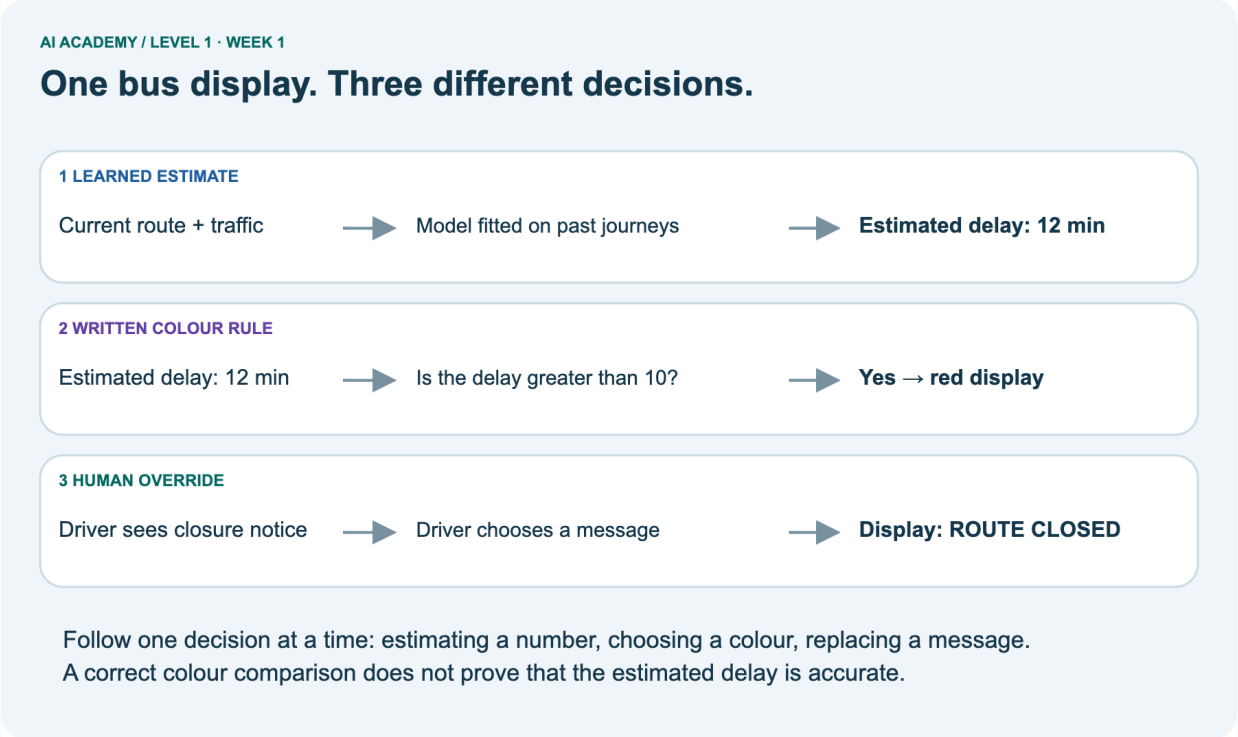
End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach the three mechanisms clearly

> Keep every explanation attached to the specific decision and the evidence supporting it.

Begin with the bus display



Follow the labelled picture with the matching explanation. Open the vector diagram

The school bus display says DELAY **12 MINUTES** in red. Leila calls it AI because it predicted the delay. Arun calls it a simple rule because the number turned red.

Both have noticed a real part of the product, but neither has explained the whole system. To investigate, ask which decision you are talking about and what **evidence** reveals how that decision was made.

An **output** is what a system produces. A **mechanism** is how it produces that **output**. Seeing a red number establishes its colour; it does not reveal where the number came from.

Similar **output**s can come from different **mechanism**s, and one product can contain several **mechanism**s.

For this lesson we have an engineering note: the delay estimator was fitted using records of earlier bus journeys; a person wrote the colour instruction; and the driver can replace the message.

These are supplied facts for our classroom case. In a real investigation, we would need trustworthy documentation, a demonstration of the **process** or another source of **mechanism evidence**.

Explain the information path

1. Input — information entering

INPUT a command, signal, photo, sound, or number

2. Process — the deciding step

PROCESS a human choice, fixed instruction, or learned pattern

3. Output — the result

OUTPUT movement, calculation, action, or **prediction**

In the first row, **fitting** means adjusting a model using examples. Applying the fitted model to a new journey is a different step. The model can produce an incorrect estimate even if it was trained carefully.

In the second row, a written threshold compares the estimate with 10. It can follow its instructions perfectly while using a bad estimate. In the third row, a person makes the message decision.

Model: claim → evidence → limited conclusion

Question: Does the red display prove that a **learned model** was used? First separate the claim from the **observation**. The **observation** is that the displayed number is red. The claim concerns the **process** that produced the number.

Next find relevant **evidence**. The engineering note says earlier journeys were used to fit the delay estimator. That note supports a learned-model classification for the estimator. The red colour alone does not.

Claim	Relevant evidence	Conclusion
The estimator uses a learned model	The note describes fitting on past journeys	Supported for the estimator
The red colour proves learning	Only the displayed colour is observed	Not supported by colour alone
The driver override is a model prediction	A driver selects the replacement message	Contradicted in this case

A strong answer states its limit: I can identify these three **mechanism**s from the supplied note, but I cannot infer the model's accuracy from one estimate.

Even if the bus arrives exactly **12 minutes** late, one success does not establish that the estimator will always work.

Prevent the “all rules are non-AI” shortcut

This course concentrates on modern machine learning. AI also includes symbolic and rule-based approaches. A simple automatic switch or robot body is not, by itself, **evidence** of AI.

Discuss how the system reasons or learns, rather than judging appearance.

AI is a broader field than learning from examples. It also includes approaches that represent knowledge and reason with rules. Therefore, do not use the shortcut that every rule is non-AI.

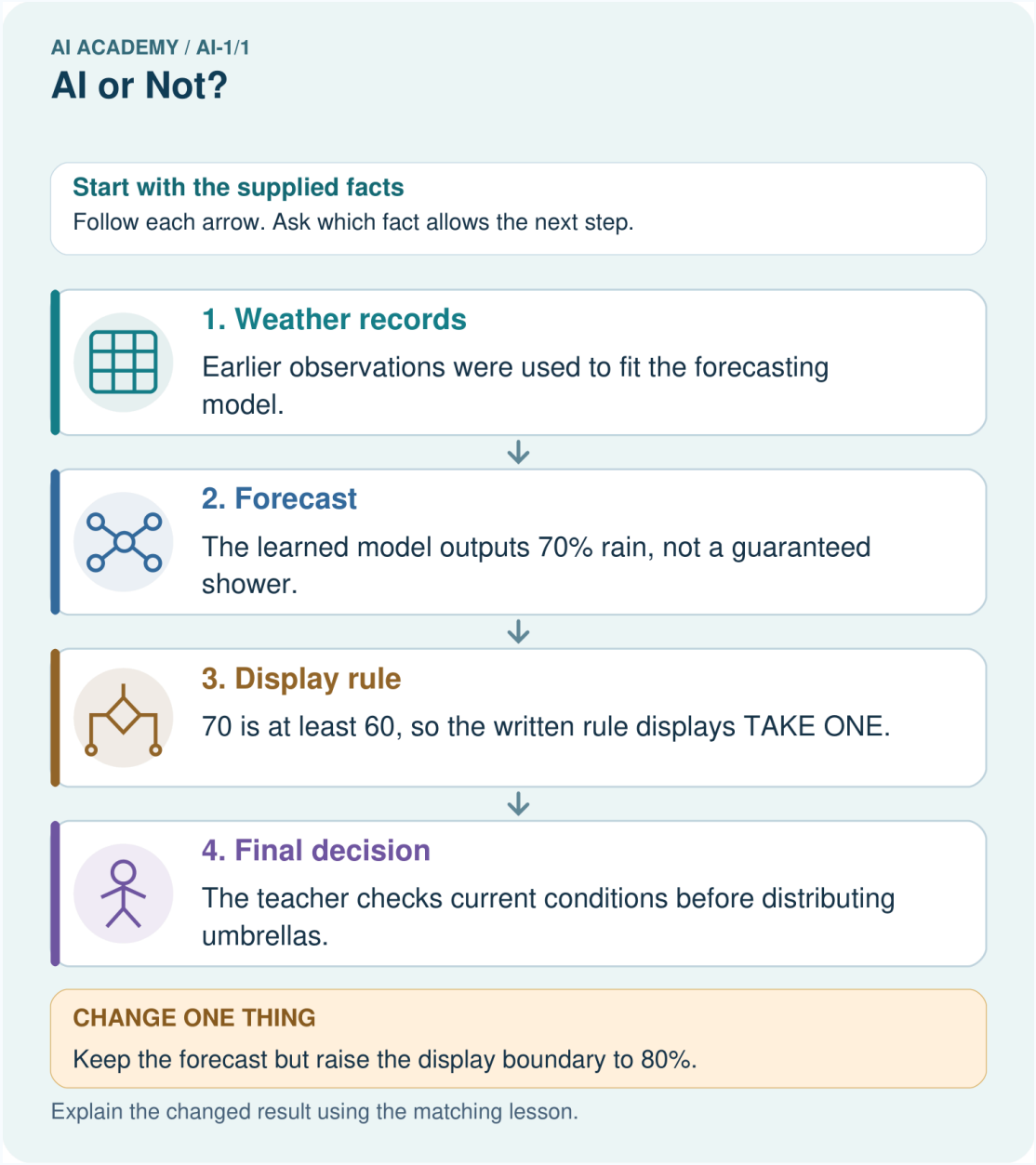
In this case, a simple colour switch is ordinary automation; identifying that switch does not classify every other part of the bus system.

Transfer the explanation to umbrellas

A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?

A weather model was fitted on earlier weather records; its forecast is **70%** rain.

A **written rule** shows TAKE ONE at rain forecasts of **60%** or more. The teacher decides whether to distribute umbrellas.



Follow the labelled picture with the matching explanation. Open the vector diagram

The **fitting** record supports calling the forecasting stage learned. Comparing 70 with 60 is an explicit rule. The final action belongs to a person.

Merely seeing TAKE ONE would not reveal any of these **mechanism**s; the **process** notes supply that **evidence**.

Change the rule, not the forecast

1. Change: Keep the forecast but raise the display boundary to **80%**.
2. **Prediction**: The display no longer says TAKE ONE.
3. Reason: 70 is below 80; changing a display rule does not refit the model.

Respect the limit

A correct rule comparison does not establish that rain will occur or that **70%** forecasts are followed by rain on about 70 out of 100 comparable days.

Use the picture deliberately

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Six-device key, vocabulary and additional explanation prompts

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

CORE IDEA Look for **evidence** of how the system reasons or learns, not just its label or screen.

UNDER THE HOOD A system should be classified by its decision **mechanism**. **Human control**, **fixed automation**, and model-based **prediction** can produce similar visible behavior, so appearance and marketing words are weak **evidence**.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

CASE A remote-control car, calculator, motion light, smart speaker, photo sorter, and robot vacuum may all look clever, but they do not make decisions in the same way.

STEP 1 — OBSERVE a command, signal, photo, sound, or number

STEP 2 — TRACE a human choice, fixed instruction, or learned pattern

STEP 3 — CONCLUDE movement, calculation, action, or **prediction**

MODEL REASONING Follow the information and identify who or what makes the important choice. A robot body, screen, voice, or the word smart cannot prove AI.

If two **mechanism**s fit the **evidence**, say that **more information** is needed.

DO NOT OVERCLAIM Do not call a system AI because it looks, moves, or talks like a person.

What new **evidence** could change this answer?

Teacher precision note

Concrete teaching case

Six devices are a motion-triggered lamp, a trained photo sorter, a remote-controlled car, a calculator, an unspecified smart speaker and an unspecified robot vacuum.

Classify all six using **mechanism evidence**. Which two need **more information**?

Exact explanation to model

The lamp uses a fixed rule; the photo sorter uses a **learned model**; the car follows human commands; the calculator performs arithmetic. The speaker and vacuum need **mechanism** details. Products can combine **mechanism**s.

A robot body or smart label proves none of them.

Explain the added worked example

Key terms

Term	Meaning
Mechanism	The process that produces a result
Written rule	An instruction specified by a person before the system applies it
Learned model	A system whose adjustable values were fitted using examples

Term	Meaning
Evidence	Information that supports or challenges a particular claim
Hybrid system	A product that combines more than one kind of process

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: AI or Not?

The facts we will use

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Weather records	Earlier observations were used to fit the forecasting model.
2. Forecast	The learned model outputs 70% rain, not a guaranteed shower.
3. Display rule	70 is at least 60, so the written rule displays TAKE ONE.
4. Final decision	The teacher checks current conditions before distributing umbrellas.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Model, then remove the help

> Use the photo-library question to check whether learners can transfer the bus explanation.

Prompt, then listen

Completion task: Use the photo-library case from the lesson. Classify the **prediction**, album-placement rule and owner move. A bicycle photo is in the wrong album. Name the first two records you would inspect.

Reasoning to listen for

Teacher check: **Prediction** is learned because it was fitted using labelled pictures. Album placement is a **written rule**; the owner move is **human control**. Inspect the stored **prediction** and the placement or move record.

A wrong album alone cannot distinguish **prediction** error from a later action.

The case-specific table above is a focused two-session sequence (90 minutes). Allow additional sessions for the rest of the weekly work; the allocations are planning examples, not validated durations.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Evidence** and decision **mechanism**s. Proposed explanation: “All rules are outside AI”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Explain that AI also includes symbolic reasoning. In the specific bus case, the simple colour threshold is automation; the full system contains a fitted predictor.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the umbrella picture as a quick check

Point to the learned stage. Does the changed display prove the weather forecast changed?

The forecast stage is learned; it remains **70%**. Only the boundary and resulting message changed.

Changed-case reasoning: The display no longer says TAKE ONE. 70 is below 80; changing a display rule does not refit the model.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep the dialogue focused

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier board sequence and alternative case-specific schedule

Demonstrate and guide practice

Board sequence

Write the given facts: Six devices are a motion-triggered lamp, a trained photo sorter, a remote-controlled car, a calculator, an unspecified smart speaker and an unspecified robot vacuum.

Write the question to explain: Classify all six using **mechanism evidence**. Which two need **more information**?

Reveal after **prediction**s: The lamp uses a fixed rule; the photo sorter uses a **learned model**; the car follows human commands; the calculator performs arithmetic. The speaker and vacuum need **mechanism** details.

Products can combine **mechanism**s. A robot body or smart label proves none of them.

Optional later transfer practice: A toy follows your steering buttons. A photo app learned from labelled pictures. A talking clock follows a written timetable. Which uses a **learned model**? What **evidence** supports each classification? Keep the greenhouse as the selected independent assessment.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Prediction stage is learned because labelled examples were used to fit the model; album placement follows the stated **written rule**; the owner move is **human control**. Inspect the predicted label and the album action record.

A correct answer from a speaking device does not identify the **mechanism** because recordings, rules, **learned model**s and human messages can all produce speech.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 8 minutes	Ask the clock prerequisite question. Record observed and inferred statements in separate columns. Read two examples aloud without praising a guess as a fact.
8 to 23 minutes	Read the bus case. Trace each row of the diagram with a finger. Ask which decision each row makes. Say that a product can combine mechanisms.

Stage	Teacher action
23 to 35 minutes	Model the red-display question using claim, relevant evidence and conclusion. Cover the final table row and ask students to justify it.
35 to 45 minutes	Pairs complete the photo-library explanation and W1. Listen for a distinction between a predicted label and an album action.
Session 2 start 0 to 8 minutes	Retrieve the distinction with the speaking-device exit question. Revisit the diagram if students rely on appearance.
8 to 23 minutes	Complete W2 and W3 in pairs. Each partner must offer a different possible cause of a misplaced photo or direction change.
23 to 38 minutes	Collect W4 as an independent assessment. Do not display its solution or substitute a hinted answer for an independent attempt.
38 to 45 minutes	Discuss one anonymised explanation and set W5 as extension or the next opening task. Record which prerequisite needs reteaching.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the device detective investigation

> The device cards are supported practice. Reserve the greenhouse for the independent attempt.

Organise the task

1. Complete the task: Sort six devices by **mechanism** and defend two borderline cases.

2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

All rules are outside AI: Explain that AI also includes symbolic reasoning. In the specific bus case, the simple colour threshold is automation; the full system contains a fitted predictor.

A correct **output** proves a **learned model**: Ask whether a calculator, recording or person could produce the same correct **output**. Require **evidence** about **fitting** or reasoning.

Unknown is a bad answer: Accept unknown when the learner identifies a specific missing fact and a useful way to obtain it.

Choose the next teaching action

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Original lab setup, alternative workflow and general guidance

Run the weekly evidence lab

Predict first, test honestly, and explain what the **evidence** means.

State your **prediction** before the result is known.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the six-device case as shared practice. Each pair creates a missing-information or changed-evidence case and predicts what could be concluded before checking it. Keep the greenhouse for an individual first attempt before hints or feedback; do not use it as a paired lab example.

Diagnose and reteach

CLAIM TO REPAIR Talking proves that a machine learned.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the explanation, not the label

> Use the selected greenhouse key after the first attempt. Keep the other tasks clearly identified as practice.

Selected independent case: greenhouse

Selected case: A school greenhouse estimates tomorrow's temperature using a model fitted on historical weather. A **written rule** displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent.

Identify each **mechanism** and explain why a correct CHECK PLANTS message would not prove an accurate temperature **prediction**.

The forecast is learned-model **prediction**; the message is a written threshold; vent opening is the caretaker's decision. The message may correctly follow a forecast that is itself inaccurate.

Compare the forecast with a predefined temperature **observation** at the relevant place and time before judging **prediction** accuracy.

Judge the evidence in the response

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give feedback the learner can use

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Workbook W1: observation and mechanism

A door opens when a visitor approaches. Its label says intelligent entrance. List two **observation**s and write one **mechanism** question whose answer would help classify the opening decision.

Observations include that the door opened when the visitor approached and that its label says intelligent entrance.

A useful question asks whether a written sensor threshold, a model fitted on examples or a person controls opening. The label is an observed marketing claim, not **mechanism evidence**.

Workbook W2: photo-library trace

Use the photo-library case from the lesson. Classify the **prediction**, album-placement rule and owner move. A bicycle photo is in the wrong album. Name the first two records you would inspect.

Prediction is learned because it was fitted using labelled pictures. Album placement is a **written rule**; the owner move is **human control**. Inspect the stored **prediction** and the placement or move record.

A wrong album alone cannot distinguish **prediction** error from a later action.

Workbook W3: competing explanations

Maya says a vacuum must learn because it changes direction after hitting a chair. Give two different **mechanism**s consistent with that **observation**. What **evidence** would distinguish them?

A written contact-sensor rule could turn the vacuum, or a learned navigation model could choose a turn. A person could also control it.

Documentation of the route-selection **process** or a controlled trace could distinguish these **mechanism**s. Direction change alone is insufficient.

Workbook W5: evidence-card design

Invent one product containing two **mechanism**s. State the supplied facts, draw an information path and write one claim the facts support and one they do not. Do not use real classmates' personal information.

Accept a coherent case such as a trained speech recogniser supplying text to a written keyword rule. The support must connect to supplied facts, and the unsupported claim must exceed them, such as assuming perfect recognition.

Check that the learner distinguishes the two decisions rather than merely naming two devices.

▼ Other practice keys, repeated greenhouse copy and earlier scoring option

Assess and explain the answers

Worked case answer

The lamp uses a fixed rule; the photo sorter uses a **learned model**; the car follows human commands; the calculator performs arithmetic. The speaker and vacuum need **mechanism** details. Products can combine **mechanism**s.

A robot body or smart label proves none of them.

Additional transfer practice question

A toy follows your steering buttons. A photo app learned from labelled pictures. A talking clock follows a written timetable. Which uses a **learned model**? What **evidence** supports each classification?

Reasoned practice answer

The photo app uses a **learned model** because **training** examples shaped its behaviour. The toy is human-controlled; the clock follows fixed instructions. Speech and movement alone do not establish AI.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Talking proves that a machine learned.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

The **prediction** uses a **learned model**; the colour uses a written threshold; the override is **human control**. The red display alone reveals no **training evidence**. Ask how the delay estimate was produced.

One product can combine **mechanism**s; simple automation does not define all of AI.

Explanations for the additional workbook examples

W1 Separate observations from explanations

W2 Complete the photo library explanation

W3 Challenge an unsupported conclusion

W4 Investigate a new hybrid system

A school greenhouse estimates tomorrow's temperature using a model fitted on historical weather. A **written rule** displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent.

Identify each **mechanism** and explain why a correct CHECK PLANTS message would not prove an accurate temperature **prediction**.

The forecast is learned-model **prediction**; the message is a written threshold; vent opening is the caretaker's decision. The message may correctly follow a forecast that is itself inaccurate.

Compare the forecast with a predefined temperature **observation** at the relevant place and time before judging **prediction** accuracy.

W5 Design an evidence card

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return later

> Keep one useful project record, adapt access to the explanation, and plan a changed-case return check.

Project evidence to keep

Choose whether the Helpful Classroom Sorter needs a person, fixed rule, or **learned model**; justify the smallest safe solution.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For the Helpful Classroom Sorter, annotate which decisions follow written instructions and which, if any, were fitted from examples.

Remove barriers; retain the reasoning goal

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Read the cases aloud and let students use arrows or spoken explanations. Supply the three **mechanism** labels if vocabulary is the barrier, but record that support.

Extension students should design two **mechanism**s producing the same **output** and propose **evidence** that would distinguish them.

For W4, require all three components classified with their supplied **evidence** and an explanation that a downstream rule can operate correctly on a wrong estimate. A label without supporting **evidence** is developing.

If the learner says the display proves accuracy, return to the two bus rows and ask what each row actually checks. After feedback, assess a fresh case rather than awarding independent mastery for the corrected W4 answer.

Plan a later check

Suggested return points in this level: ai-1/2, ai-1/4, ai-1/7. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Connect to Week 2

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 2: **Input**, **Process**, **Output**. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 2 follows how information is represented and changed between components. Keep the three-row bus diagram: the delay number and the warning colour are different representations of related information.

<details>
<summary>Original project and extension headings</summary>

Connect and extend learning

Helpful Classroom Sorter

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case
</details>

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> This repeats the main teaching cycle in checklist form. Use one planning route, not both.

Preparation checklist

Teaching target: Distinguish an observed result from an explanation of how it was produced

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session	Plan	Resources
Session 1 · Understand · 45 min	0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.	Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.
Session 2 · Practise · 45 min	0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.	Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

Session	Plan	Resources
Session 3 · Investigate and transfer · 45 min	0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.	Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice, support and assessment reminders

Before class, solve the supported practice: Use the photo-library case from the lesson. Classify the **prediction**, album-placement rule and owner move. A bicycle photo is in the wrong album.

Name the first two records you would inspect.

Reasoned practice answer: **Prediction** is learned because it was fitted using labelled pictures. Album placement is a **written rule**; the owner move is **human control**. Inspect the stored **prediction** and the placement or move record.

A wrong album alone cannot distinguish **prediction** error from a later action.

Check every learner before discussion: Use the worked case for **Evidence** and decision **mechanism**s. Proposed explanation: “All rules are outside AI”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Explain that AI also includes symbolic reasoning. In the specific bus case, the simple colour threshold is automation; the full system contains a fitted predictor.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Keep the forecast but raise the display boundary to **80%**. Require a **prediction** and a reason before comparing. Expected result: The display no longer says TAKE ONE.

Reason: 70 is below 80; changing a display rule does not refit the model.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and digital tools**

REFERENCE · SECTION 8

Reference: feedback and digital tools

> Use tools to keep an explanation and its evidence. This week does not require a coding lab.

Keep supported practice and assessment separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Coding guidance for later weeks

Week 1 has no Python lab. Retain this routine for later coding weeks.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

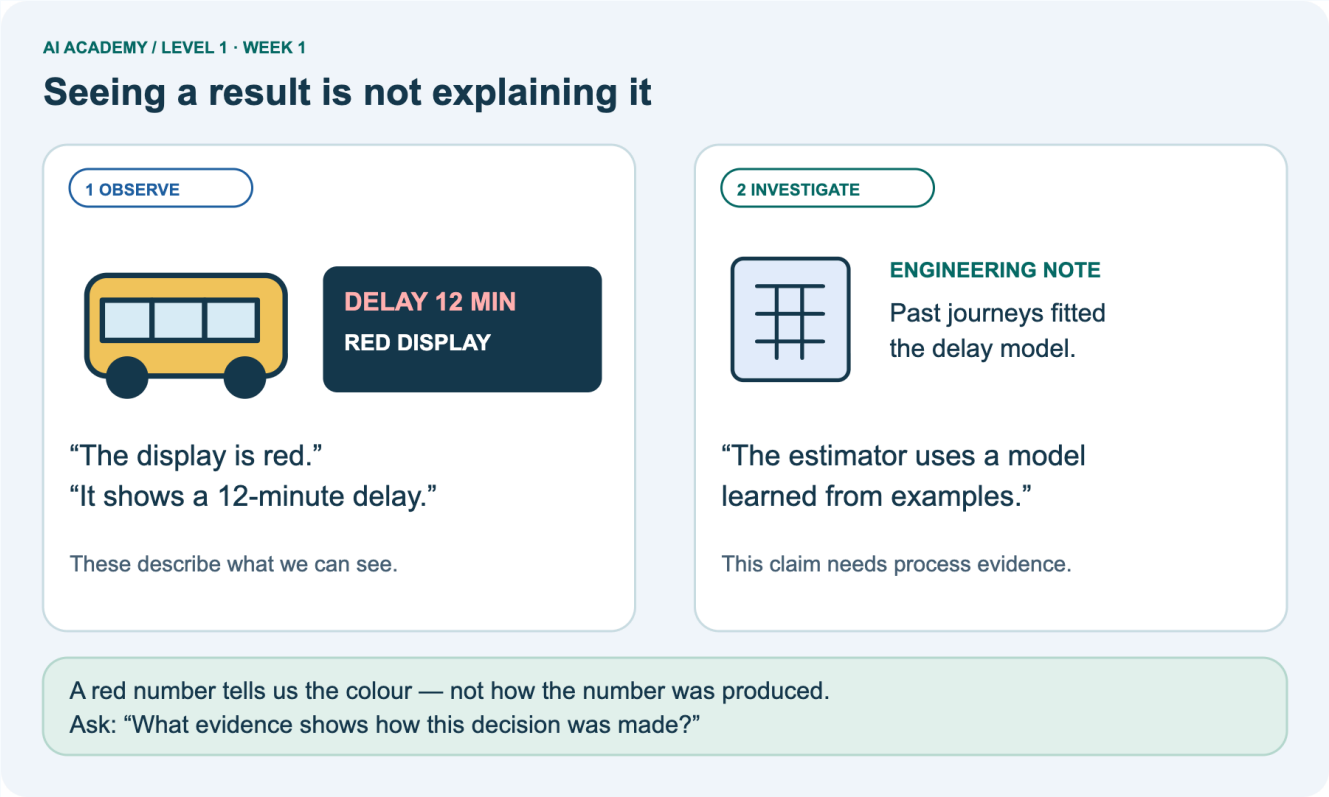
▼ Reference: teach with the six pictures

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture that serves your next teaching move rather than presenting the gallery as a second lesson.

01. Seeing a result is not explaining it



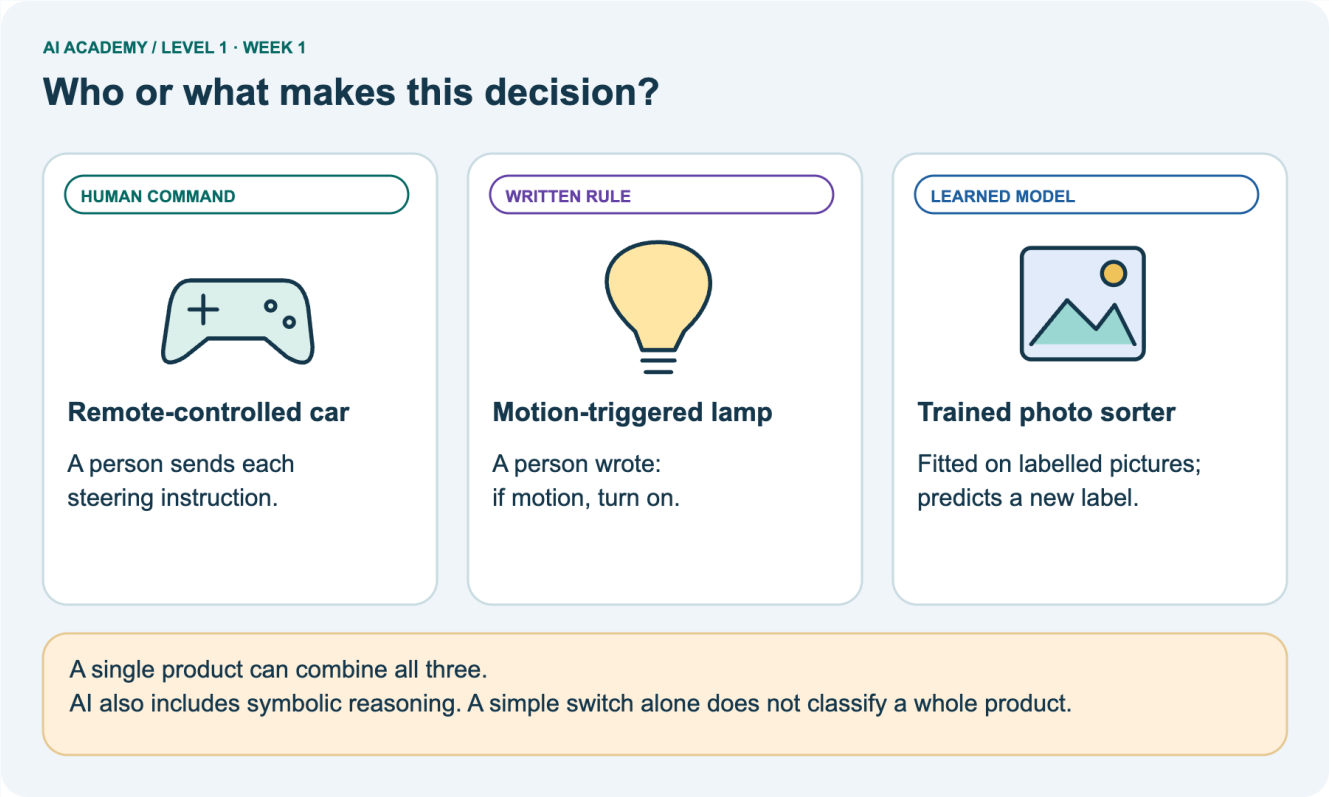
Follow the labelled picture with the matching explanation. Open the vector diagram

A bus has a red display showing a 12-minute delay. That is an **observation**. A supplied engineering note says past journeys fitted the delay model; that is **evidence** about its **mechanism**.

Name one **observation** and one claim that needs more **evidence**.

Teacher prompt: Name one **observation** and one claim that needs more **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Who or what makes this decision?



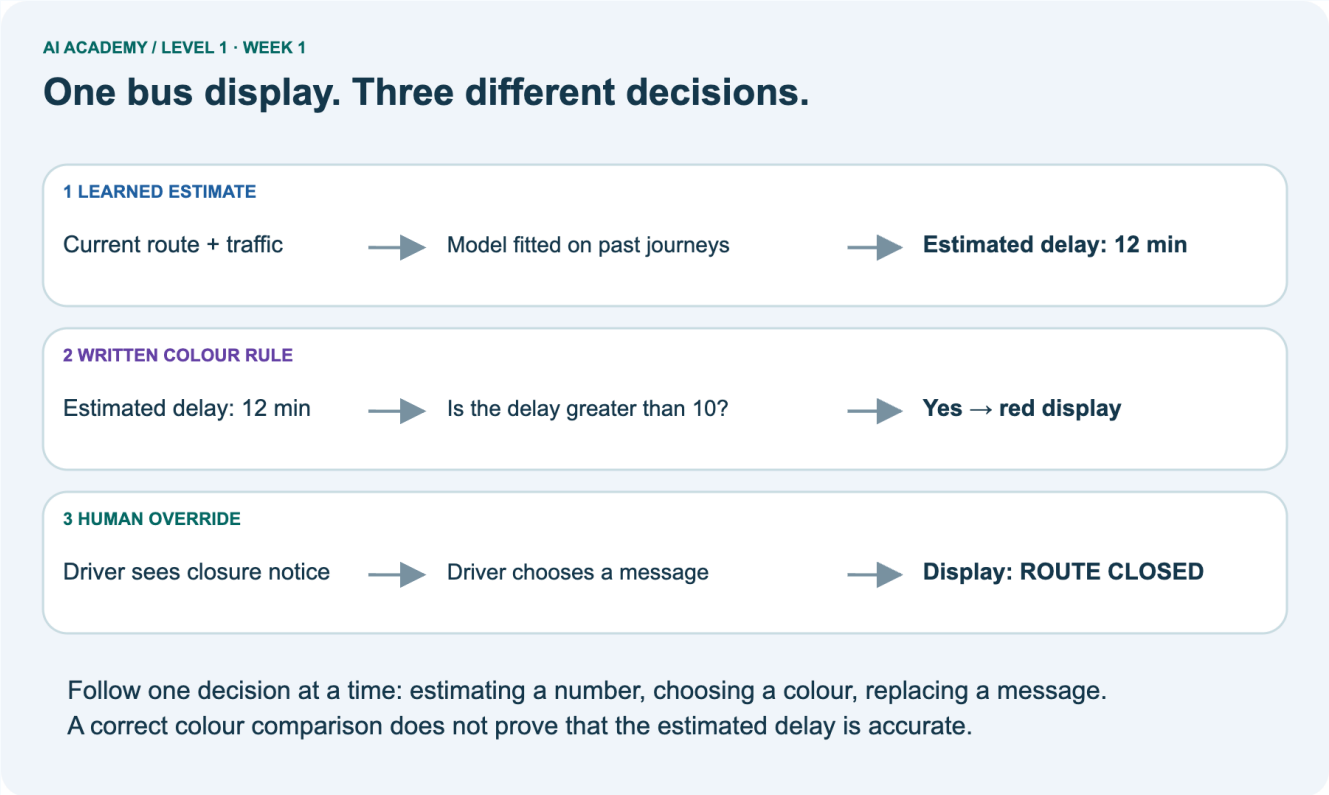
Follow the labelled picture with the matching explanation. Open the vector diagram

A remote-controlled car follows human commands, a motion lamp follows a **written rule**, and a trained photo sorter applies a **learned model**. A product can combine **mechanism**s. AI also includes symbolic reasoning.

Point to who or what makes each decision.

Teacher prompt: Point to who or what makes each decision. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. One bus display. Three different decisions.



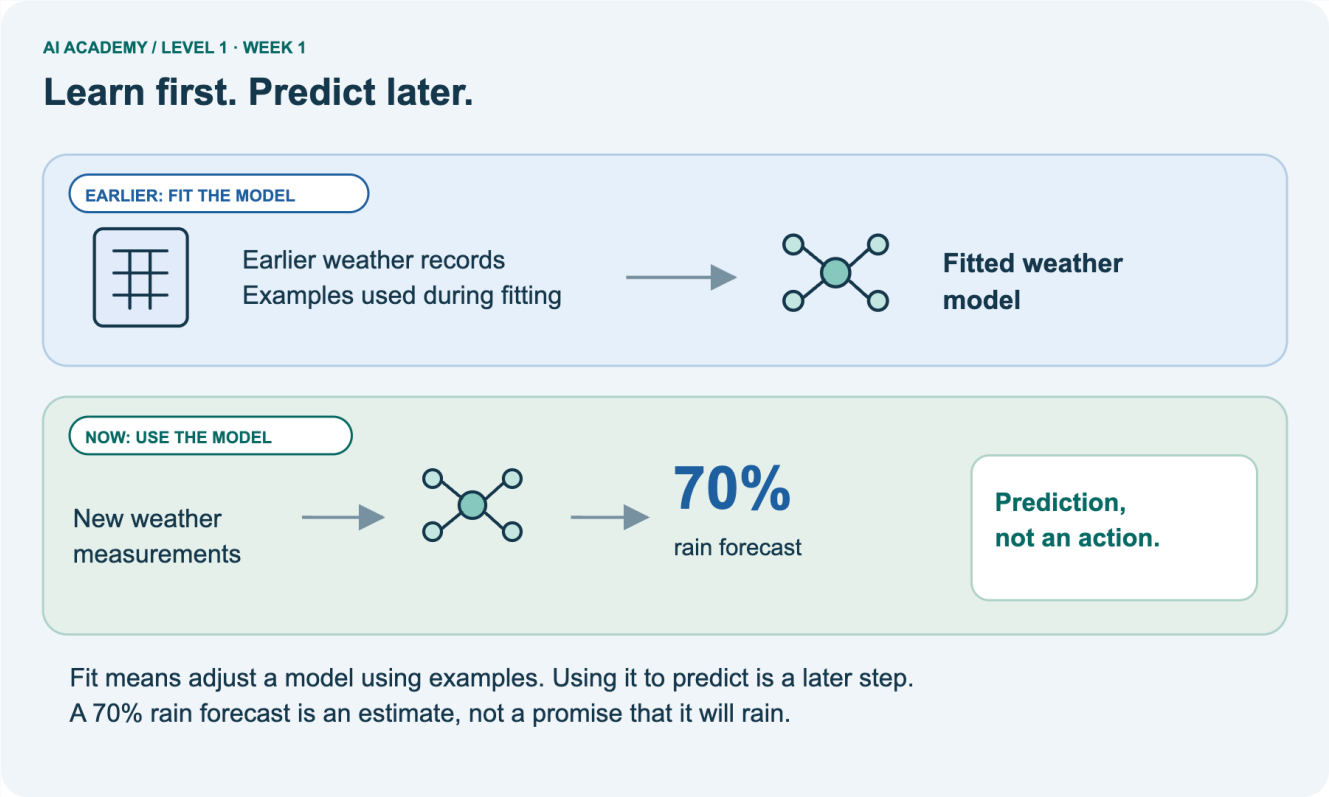
Follow the labelled picture with the matching explanation. Open the vector diagram

Three separate rows trace the bus estimate, display colour and driver override. The fitted model estimates **12 minutes**; the written greater-than-10 rule makes the display red; the driver can choose ROUTE CLOSED.

Trace the number, colour and override separately.

Teacher prompt: Trace the number, colour and override separately. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Learn first. Predict later.



Follow the labelled picture with the matching explanation. Open the vector diagram

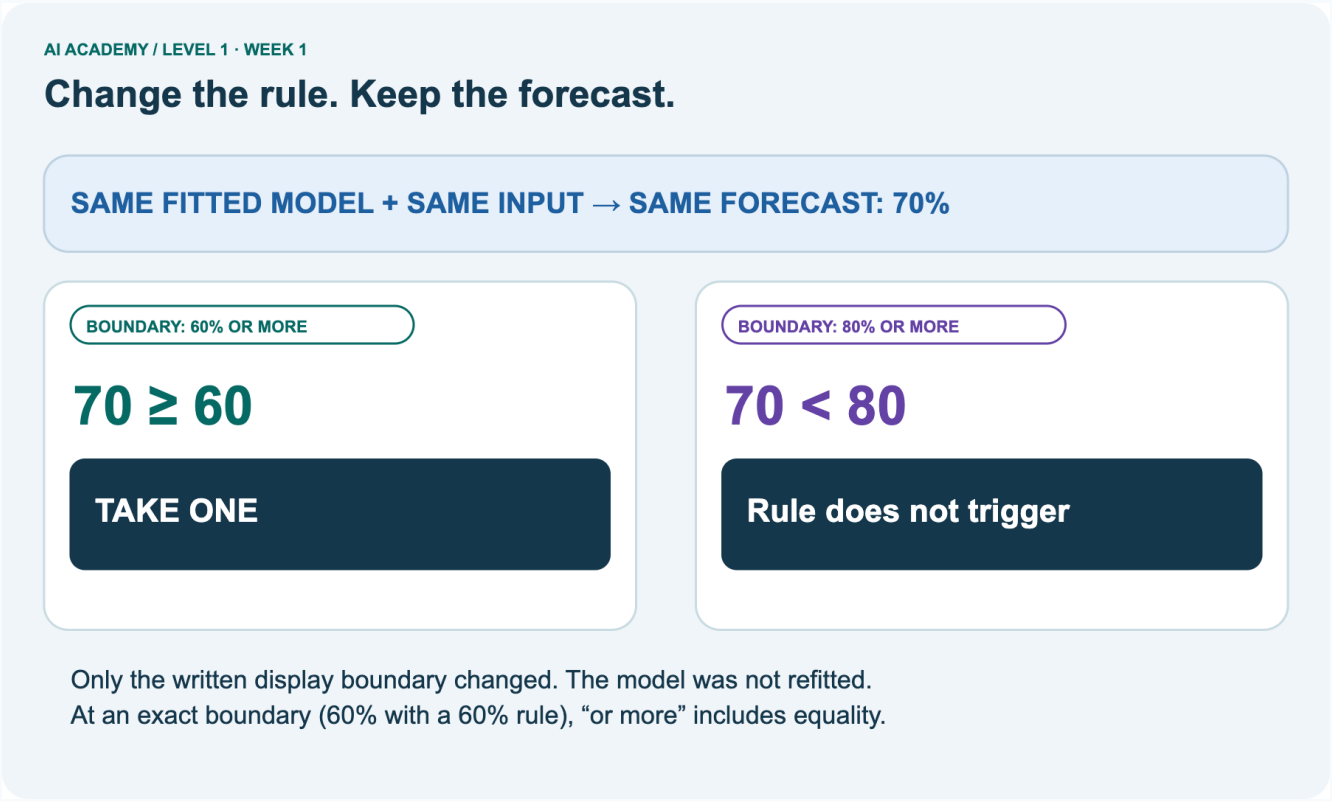
Earlier weather records are used to fit a model. Later, new measurements are passed to that fitted model to obtain a 70 percent rain forecast. It is an estimate, not a guarantee.

Which steps use earlier examples, and which use a fitted model?

Teacher prompt: Which steps use earlier examples, and which use a fitted model? Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Changed-case reasoning: 70 is below 80; changing a display rule does not refit the model.

05. Change the rule. Keep the forecast.



Follow the labelled picture with the matching explanation. Open the vector diagram

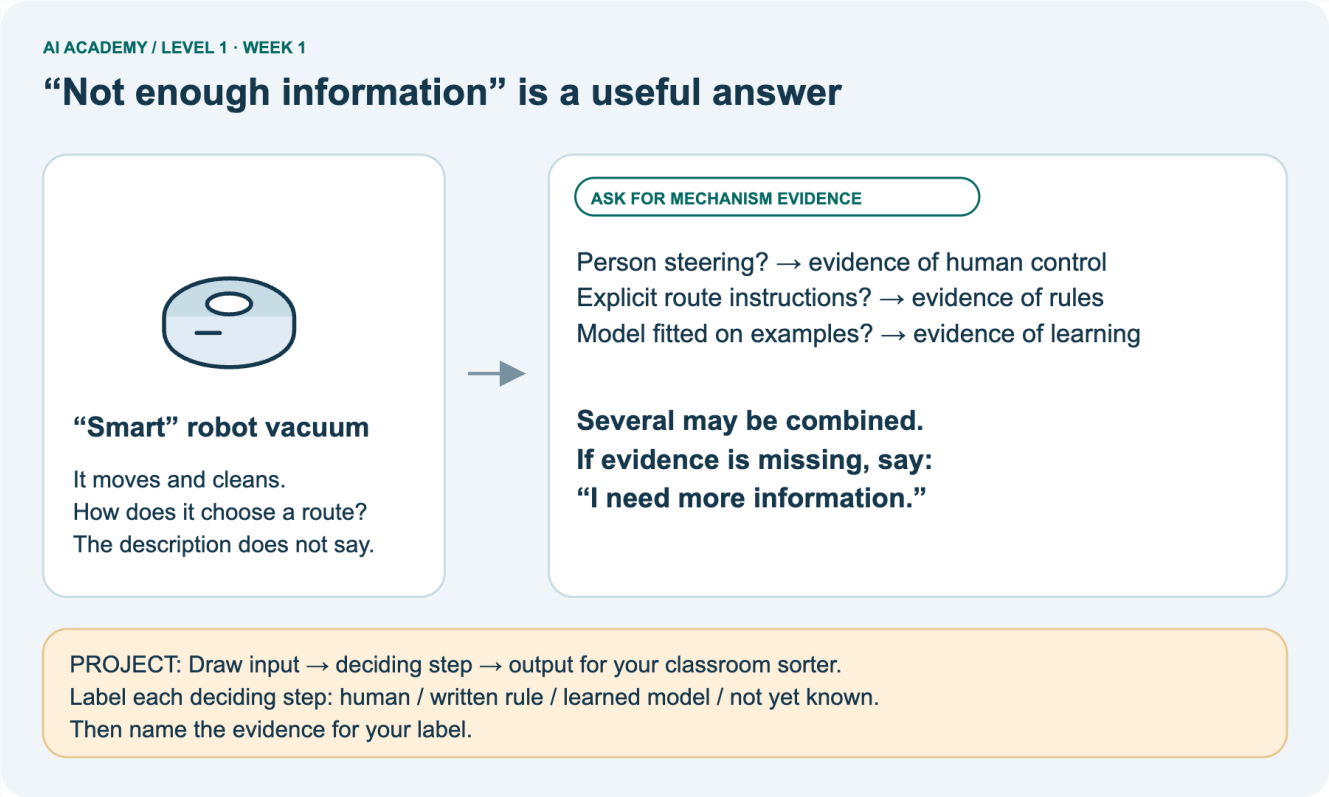
With a 70 percent rain forecast, a 60 percent-or-more rule displays TAKE ONE. An 80 percent-or-more rule does not trigger that message. Only the boundary changed; the fitted model and forecast did not.

Keep the forecast fixed at **70%**. What changes when the boundary becomes **80%**?

Teacher prompt: Keep the forecast fixed at **70%**. What changes when the boundary becomes **80%**? Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A correct rule comparison does not establish that rain will occur or that **70%** forecasts are followed by rain on about 70 out of 100 comparable days.

06. “Not enough information” is a useful answer



Follow the labelled picture with the matching explanation. Open the vector diagram

An unspecified smart robot vacuum moves and cleans, but its route **mechanism** is unknown. Ask for **evidence** of human commands, **written rule**s or model **fitting**, and allow combinations. Apply this reasoning to a classroom sorter.

Which **evidence** would change your answer about the vacuum?

Teacher prompt: Which **evidence** would change your answer about the vacuum? Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The forecast stage is learned; it remains **70%**. Only the boundary and resulting message changed.

▼ Whole-gallery teaching note

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Seeing a result is not explaining it

02. Who or what makes this decision?

03. One bus display. Three different decisions.

04. Learn first. Predict later.

05. Change the rule. Keep the forecast.

06. “Not enough information” is a useful answer

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.